Name
Date
Course

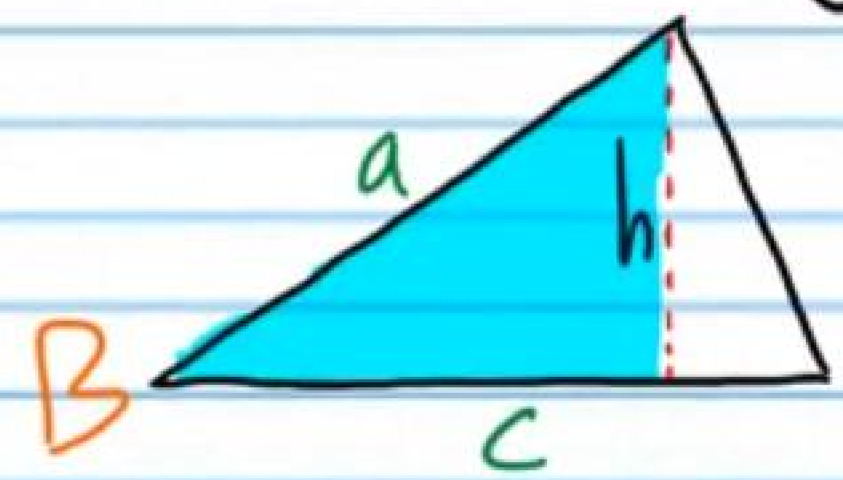
Applications of Trigonometry

Finding the Area of a Triangle with Trigonometry

The Area of an SAS Triangle

$\sin B = \frac{h}{a}$

$a \sin B = h$



$$\text{Area} = \frac{\text{base} \times \text{height}}{2}$$

$$\text{Area} = \frac{c\,a\sin B}{2} = \frac{1}{2}ac\sin B$$

The Area of an SAS Triangle

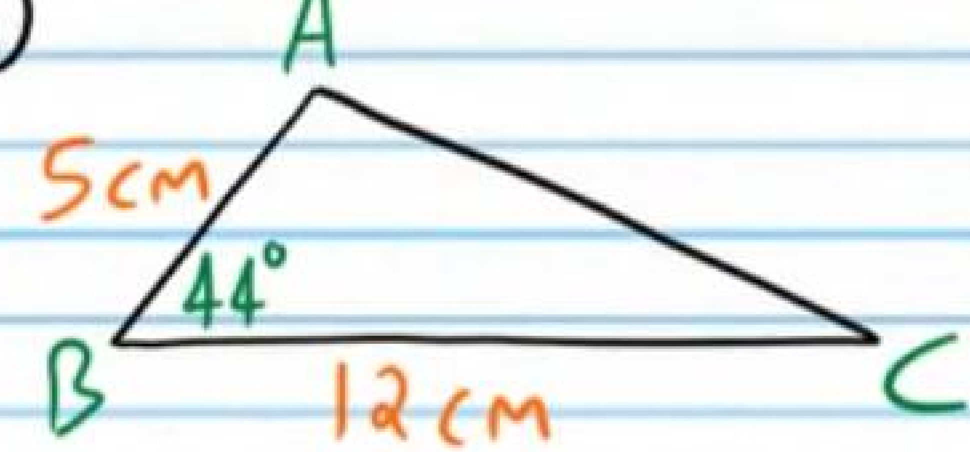
The area of a triangle is equal to the length of one side times the length of another side times the sine of the measure of the angle between the two sides divided by two.

$$\text{Area} = \frac{1}{2}ab\sin C \qquad \text{Area} = \frac{1}{2}bc\sin A$$

$$\text{Area} = \frac{1}{2}ac\sin B$$

Find the area of the triangles below.
Round to the nearest tenth.

①

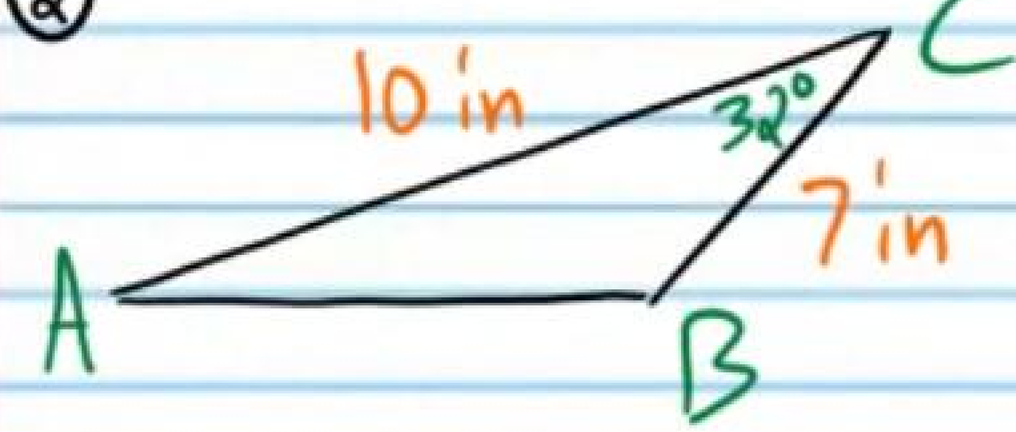


$\text{Area} = \frac{1}{2}ac\sin B$

$\text{Area} = \frac{1}{2}(12)(5)\sin 44°$

$\boxed{\text{Area} \approx 20.8\text{ cm}^2}$

②



$\text{Area} = \frac{1}{2}ab\sin C$

$\text{Area} = \frac{1}{2}(7)(10)\sin 32°$

$\boxed{\text{Area} \approx 18.5\text{ in}^2}$

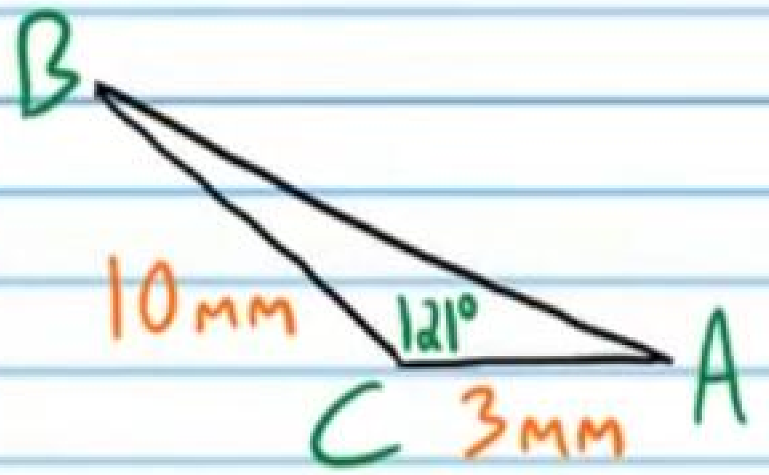
③ $b = 14\text{ ft}, c = 8.8\text{ ft}, A = 36°$

$\text{Area} = \frac{1}{2}bc\sin A$

$\text{Area} = \frac{1}{2}(14)(8.8)\sin 36°$

$\boxed{\text{Area} \approx 36.2\text{ ft}^2}$

④

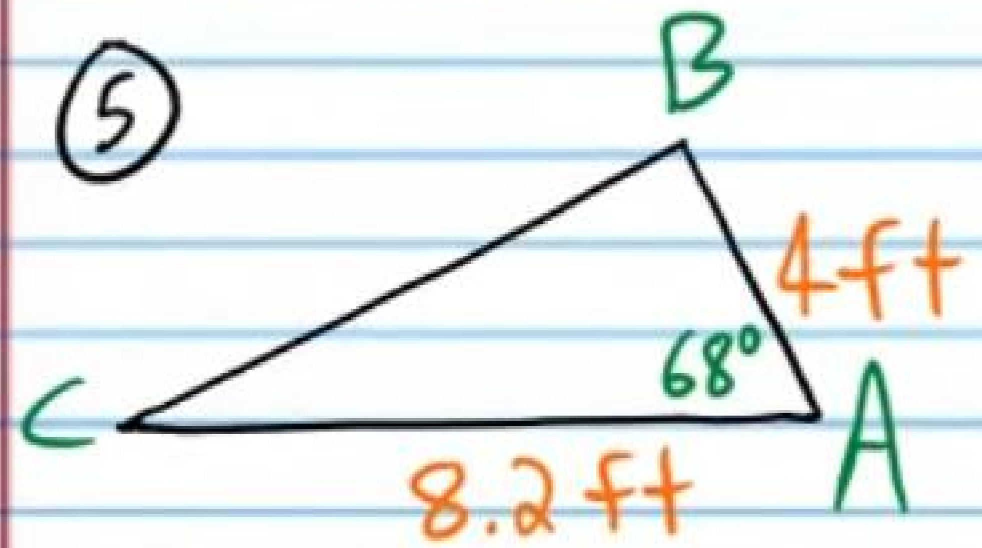


$$\text{Area} = \frac{1}{2}ab\sin C$$

$$\text{Area} = \frac{1}{2}(10)(3)\sin 121^\circ$$

$$\boxed{\text{Area} = 12.9\text{ mm}^2}$$

⑤



$$\text{Area} = \frac{1}{2}bc\sin A$$

$$\text{Area} = \frac{1}{2}(8.2)(4)\sin 68^\circ$$

$$\boxed{\text{Area} \approx 15.2\text{ ft}^2}$$

⑥ $c = 66$ km, $a = 60$ km, $B = 142^\circ$

$$\text{Area} = \frac{1}{2}ac\sin B$$

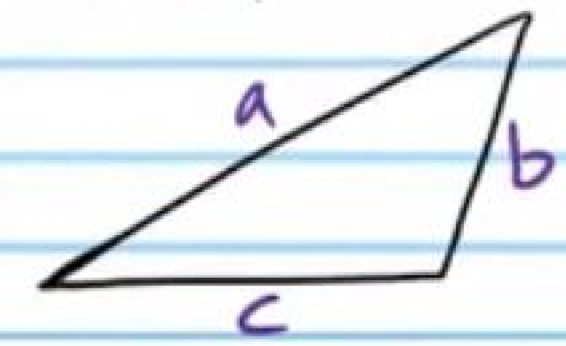
$$\text{Area} = \frac{1}{2}(60)(66)\sin 142^\circ$$

$$\boxed{\text{Area} \approx 1{,}219\text{ km}^2}$$

The Area of an SSS Triangle

Heron's Formula

If a, b, and c are the side lengths of a triangle, then the area of the triangle is equal to the expression below.



$$\text{Area} = \sqrt{s(s-a)(s-b)(s-c)}$$

where $s = \dfrac{a+b+c}{2}$

Find the area of the triangles below.
Round to the nearest hundredth.

⑦ I) $s = \dfrac{a+b+c}{2} = \dfrac{3+7+8}{2} = \dfrac{18}{2}$

$$= 9$$

II) $\text{Area} = \sqrt{s(s-a)(s-b)(s-c)}$

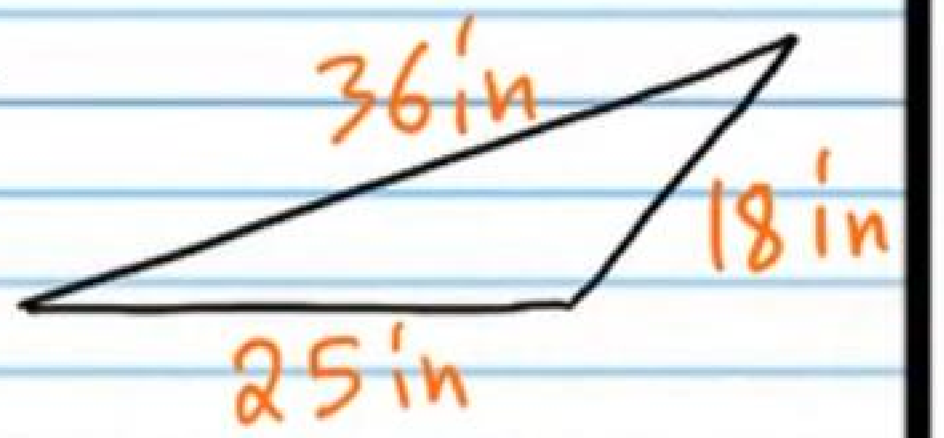
$$= \sqrt{9(9-3)(9-7)(9-8)}$$

$$= \sqrt{9(6)(2)(1)}$$

$$= \sqrt{108}$$

$$= \boxed{10.39\text{ ft}^2}$$

⑧ I) $s = \frac{a+b+c}{2} = \frac{36+18+25}{2}$



$= \frac{79}{2}$

$= 39.5$

II) $\text{Area} = \sqrt{s(s-a)(s-b)(s-c)}$

$= \sqrt{39.5(39.5-36)(39.5-18)(39.5-25)}$

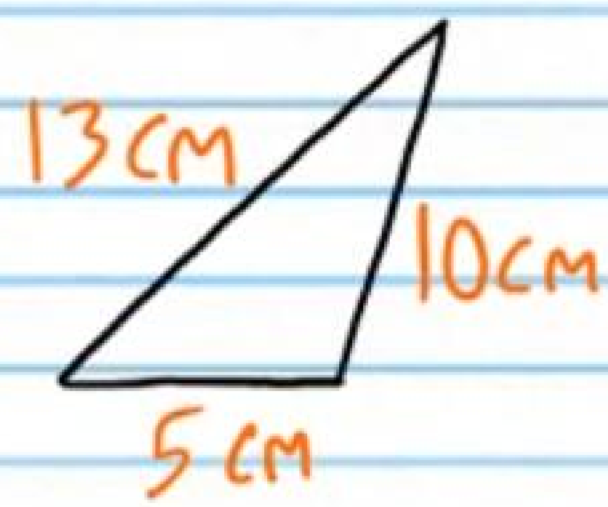
$= \sqrt{39.5(3.5)(21.5)(14.5)}$

$= \sqrt{43{,}099.4375}$

$\approx \boxed{207.6 \text{ in}^2}$

⑨ I) $s = \frac{a+b+c}{2} = \frac{13+10+5}{2} = \frac{28}{2}$

$= 14$



II) $\text{Area} = \sqrt{s(s-a)(s-b)(s-c)}$

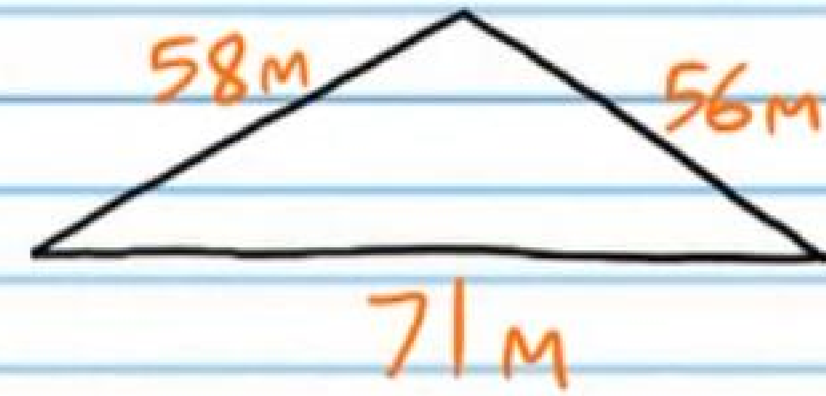
$= \sqrt{14(14-13)(14-10)(14-5)}$

$= \sqrt{14(1)(4)(9)}$

$= \sqrt{504}$

$\approx \boxed{22.45 \text{ cm}^2}$

⑩ I) $s = \frac{a+b+c}{2} = \frac{58+56+71}{2}$



$= \frac{185}{2}$

$= 92.5$

II) $\text{Area} = \sqrt{s(s-a)(s-b)(s-c)}$

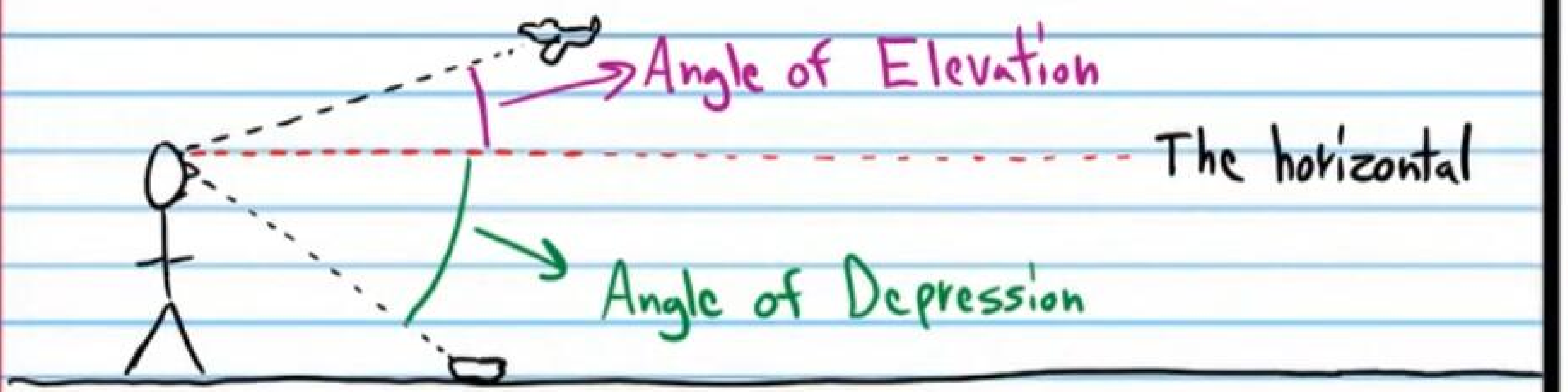
$= \sqrt{92.5(92.5-58)(92.5-56)(92.5-71)}$

$= \sqrt{92.5(34.5)(36.5)(21.5)}$

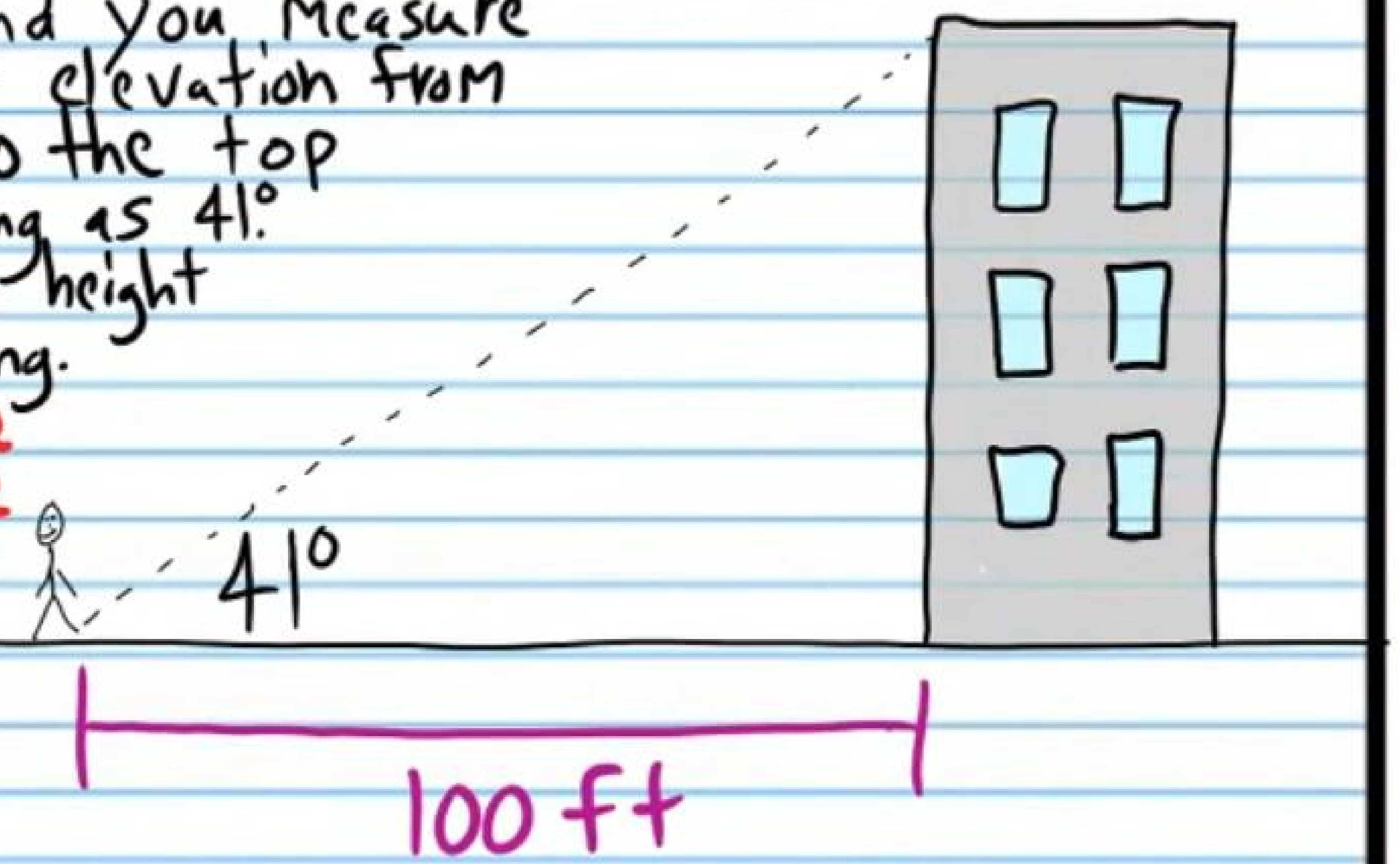
$= \sqrt{2{,}504{,}333.438}$

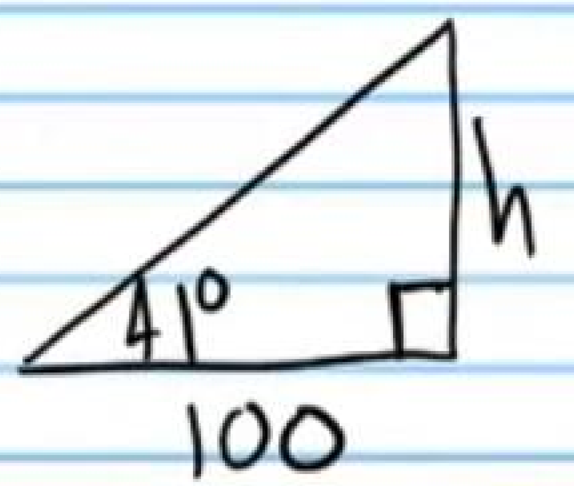
$\approx \boxed{1{,}582.51 \text{ m}^2}$

Angle of Elevation and Angle of Depression



(11) You stand 100 feet from a building and you measure the angle of elevation from the ground to the top of the building as 41°. Determine the height of the building.
Round to the nearest whole number.



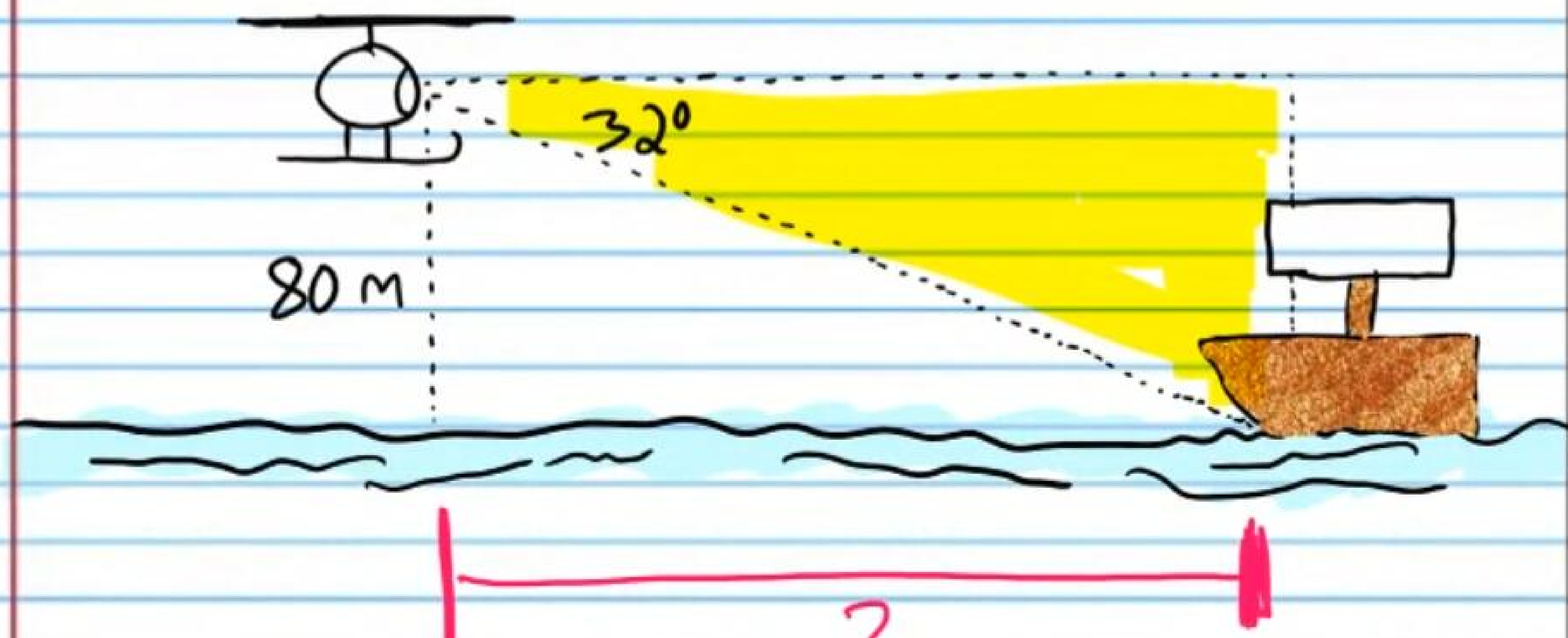


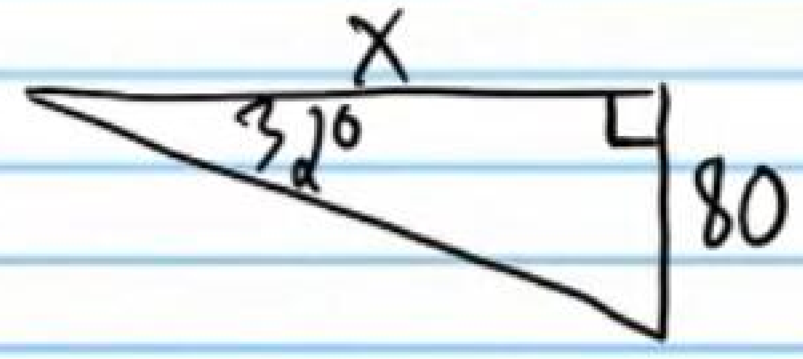
$$\tan 41° = \frac{h}{100}$$

$$100 \tan 41° = h$$

$$\boxed{h \approx 87 \text{ ft}}$$

(12) The angle of depression from a helicopter to a ship is 32°. If the pilot knows that he is flying 80 meters above the water, what is the horizontal distance from the helicopter to the ship?
Round to the nearest whole number.





$$\tan 32° = \frac{80}{x}$$

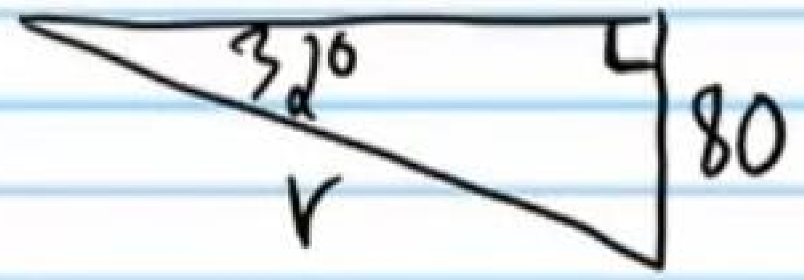
$$x \cdot \tan 32° = \frac{80}{x} \cdot x$$

$$x \tan 32° = 80$$

$$\frac{x \tan 32°}{\tan 32°} = \frac{80}{\tan 32°}$$

$$x = \frac{80}{\tan 32°} \approx \boxed{128 \text{ m}}$$

(13) For problem #12, what is the length of the shortest path from the helicopter to the ship? Round to the nearest whole number.



$$\sin 32^\circ = \frac{80}{r}$$

$$r \cdot \sin 32^\circ = \frac{80}{r} \cdot r$$

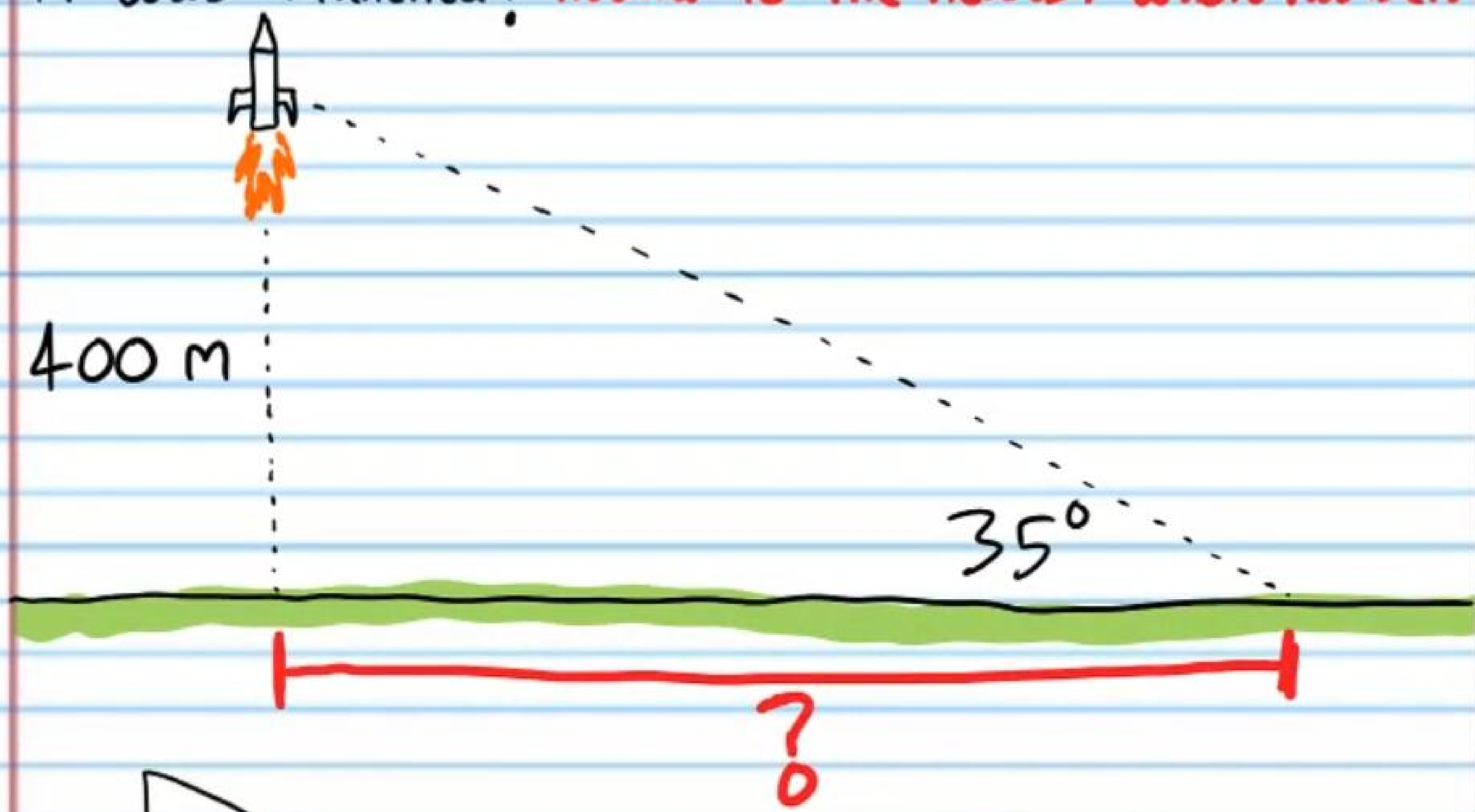
$$r \sin 32^\circ = 80$$

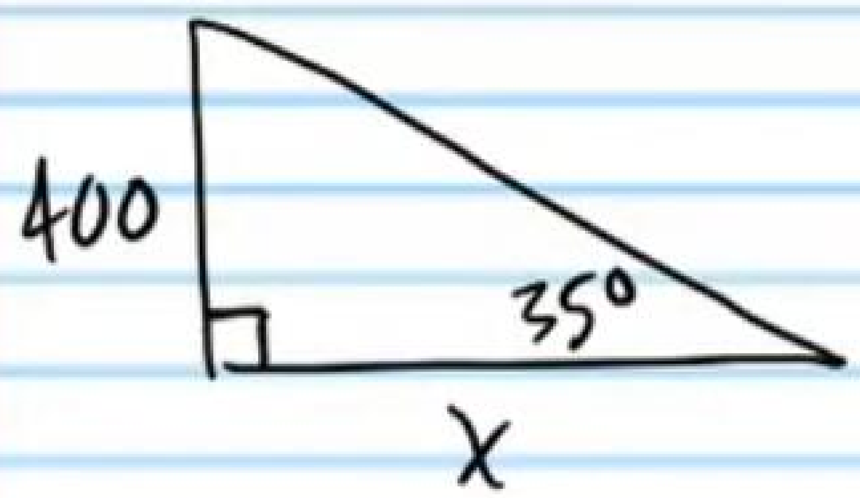
$$\frac{r \sin 32^\circ}{\sin 32^\circ} = \frac{80}{\sin 32^\circ}$$

$$r = \frac{80}{\sin 32^\circ}$$

$$\boxed{r \approx 151 \text{ m}}$$

(14) Some kids launch a toy rocket at the park. After one second in the air, its altitude is 400 meters. If the angle of elevation from the ground where the children are sitting is 35°, how far did the kids move from the rocket before it was launched? Round to the nearest whole number.





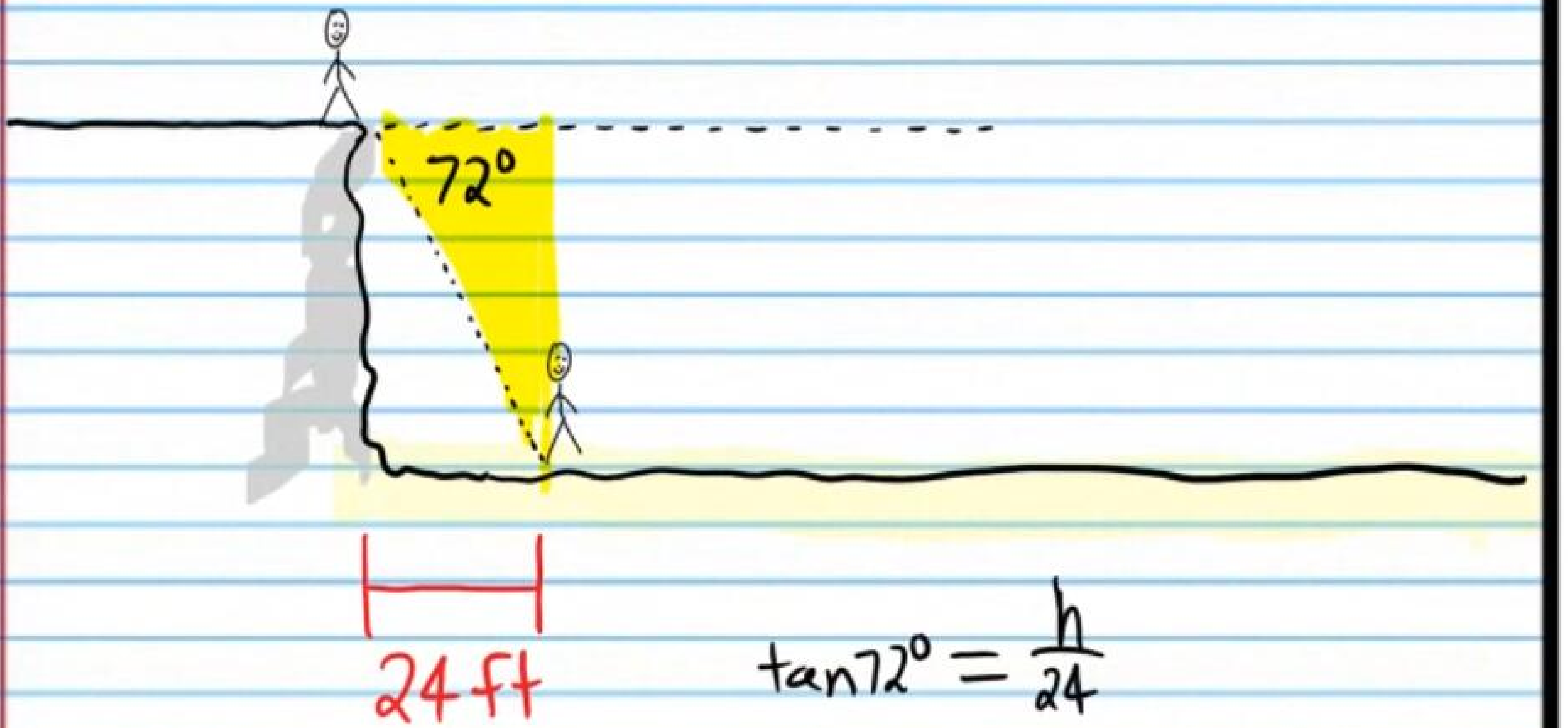
$$\tan 35^\circ = \frac{400}{x}$$

$$x = \frac{400}{\tan 35^\circ}$$

$$\boxed{x \approx 571 \text{ m}}$$

(15) Steve is on the edge of a cliff looking down to Joe. The angle of depression from the edge of the cliff to Joe is 72°. If Joe is 24 feet from the base of the cliff, how high is the cliff?

Round to the nearest whole number.

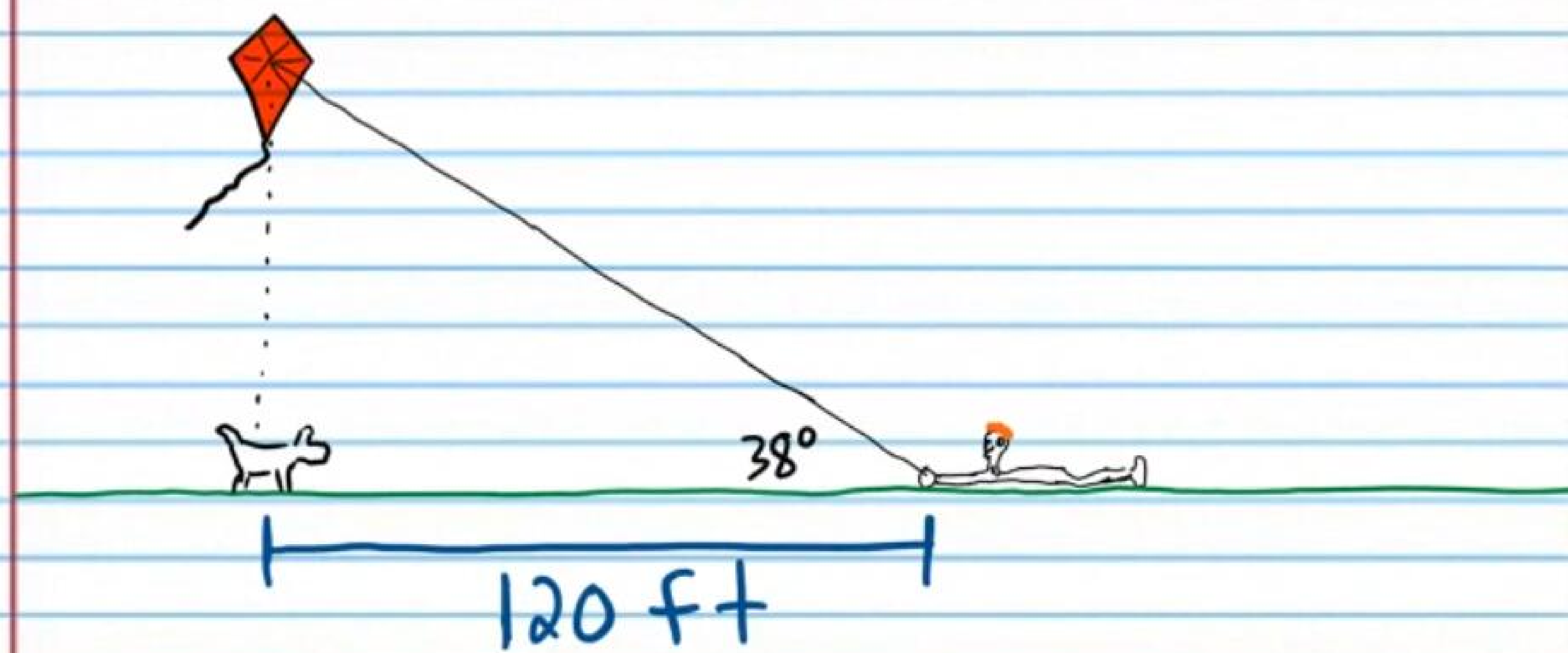


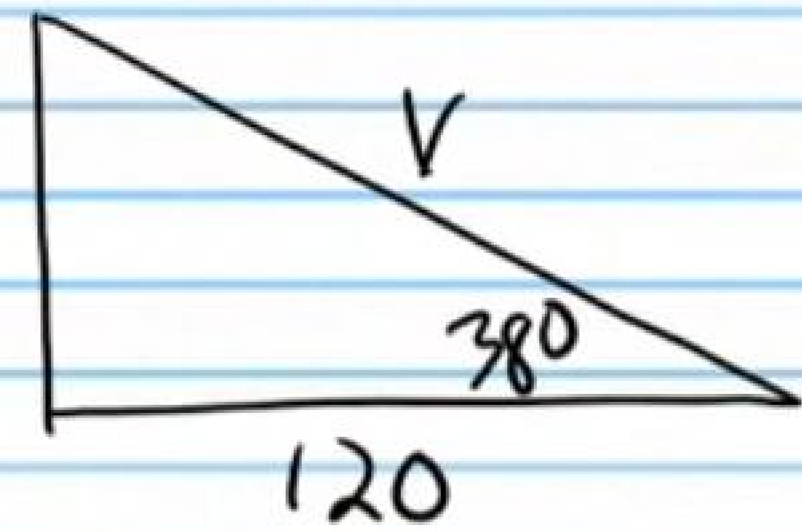
$$\tan 72^\circ = \frac{h}{24}$$

$$24\tan 72^\circ = h$$

$$\boxed{h \approx 74\text{ ft}}$$

(16) Ryan's kite is flying directly above his dog Coco. If Coco is 120 feet from Ryan, what is the length of the string that Ryan let out to raise the kite? The angle of elevation from Ryan's hands to the kite is 38°. **Round to the nearest whole number.**





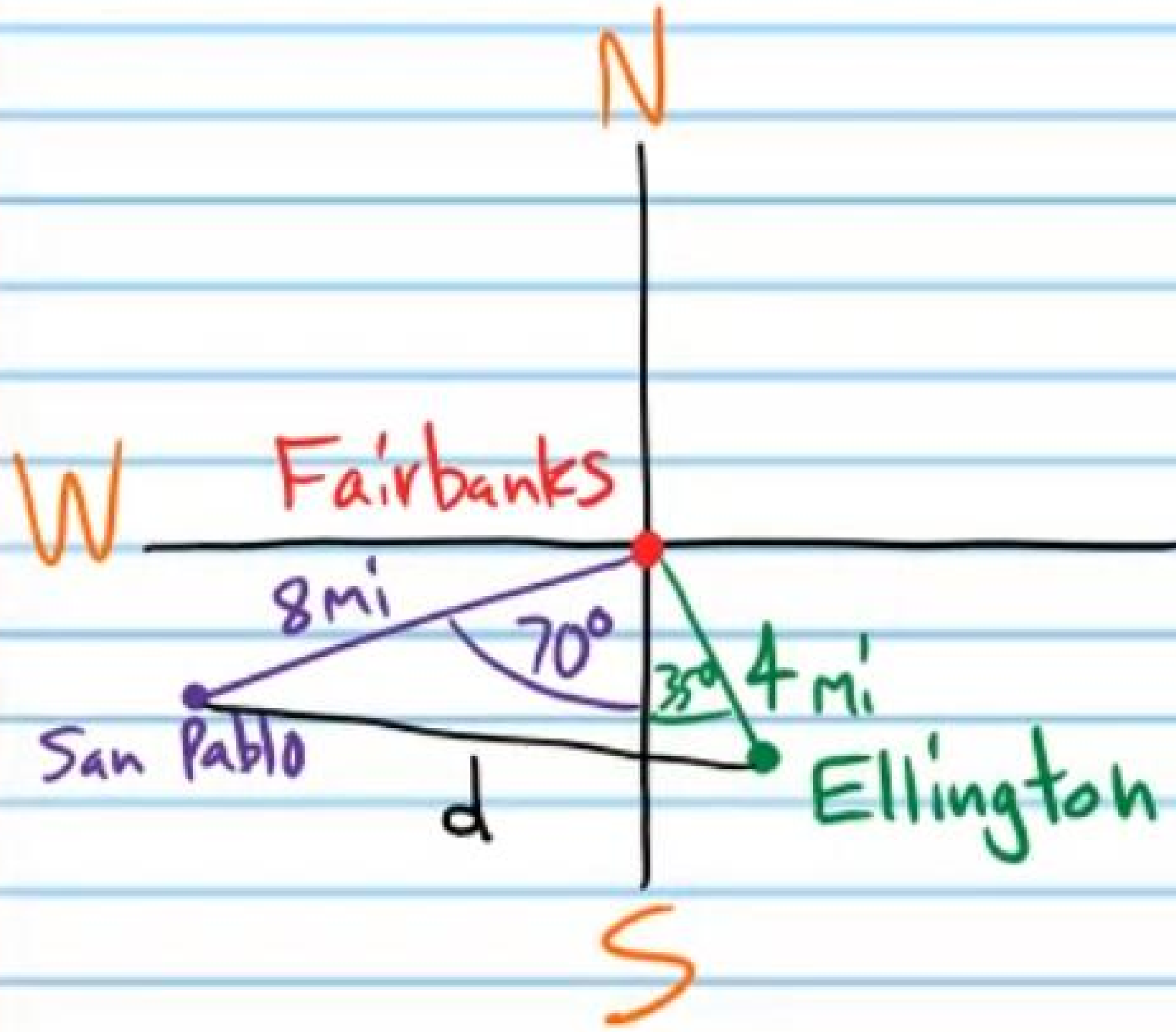
$$\cos 38^\circ = \frac{120}{r}$$

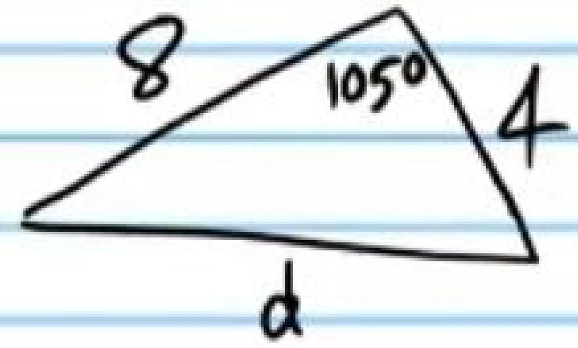
$$r = \frac{120}{\cos 38^\circ}$$

$$\boxed{r \approx 152\text{ ft}}$$

North-South-East-West Problems

(17) A map shows that a person in the city of Fairbanks must hike 4 miles on a heading 35 degrees east of south (S35°E) to arrive in the town of Ellington. If a person wants to hike from Fairbanks to the town of San Pablo, he must walk 8 miles on a heading 70 degrees west of south (S70°W). How far apart are the towns of Ellington and San Pablo?





$$c^2 = a^2 + b^2 - 2ab\cos C$$

$$d^2 = 8^2 + 4^2 - 2(8)(4)\cos 105°$$

$$d^2 = 64 + 16 - 64\cos 105°$$

$$d^2 = 80 - 64\cos 105°$$

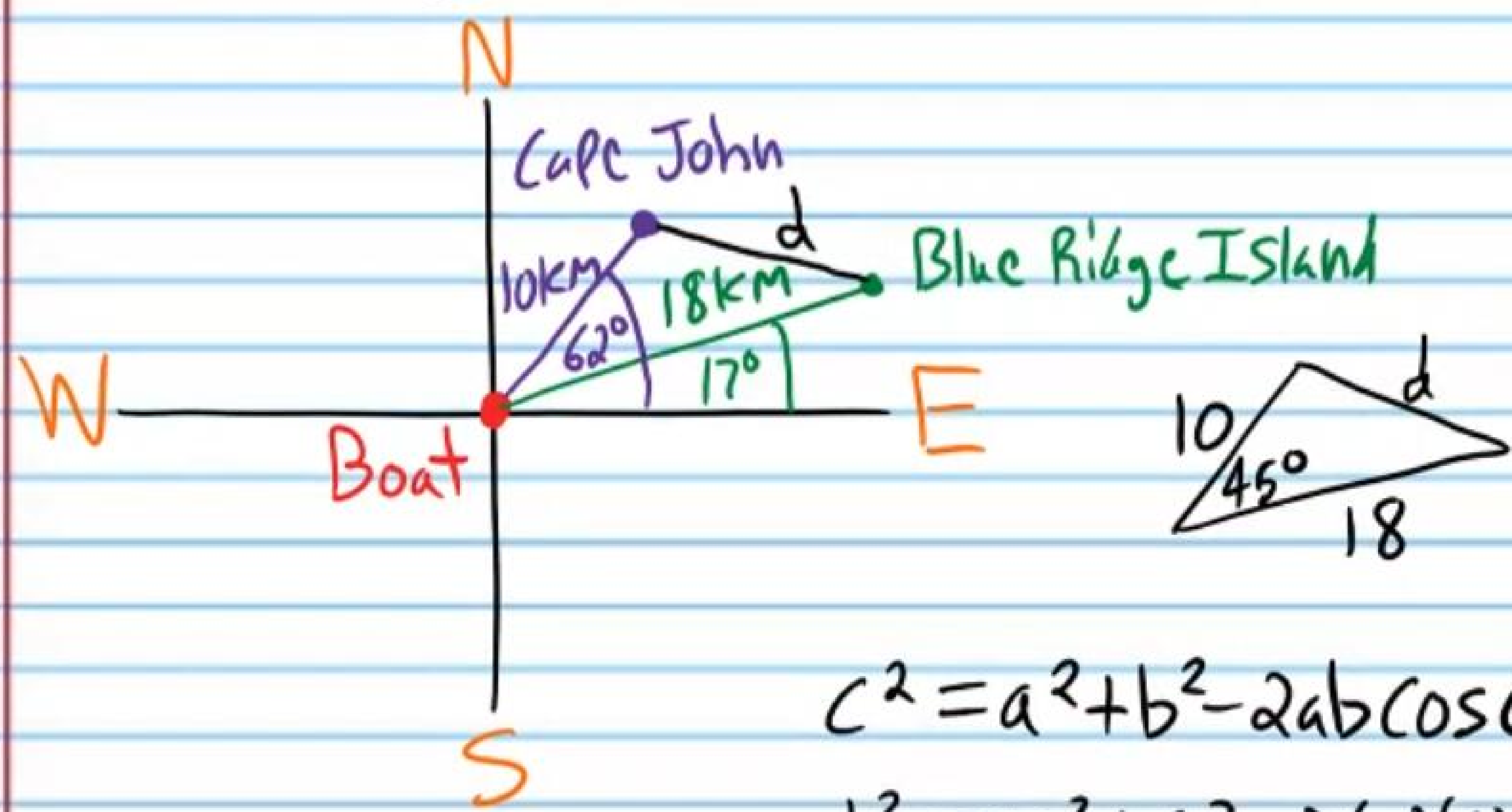
$$d^2 \approx 80 - (-16.56\ldots)$$

$$d^2 \approx 96.564\ldots$$

$$\sqrt{d^2} \approx \sqrt{96.564\ldots}$$

$$\boxed{d \approx 9.83 \text{ mi}}$$

(18) A gps device (global positioning system) shows that a boat must move 18 kilometers on a bearing 17 degrees north of east (E17°N) to get to Blue Ridge Island. The device also shows that the same boat must move 10 kilometers on a bearing 62 degrees north of east (E62°N) to get to Cape John. How far apart are Blue Ridge Island and Cape John?



$$c^2 = a^2 + b^2 - 2ab\cos C$$

$$d^2 = 10^2 + 18^2 - 2(10)(18)\cos 45°$$

$$d^2 = 100 + 324 - 360\cos 45$$

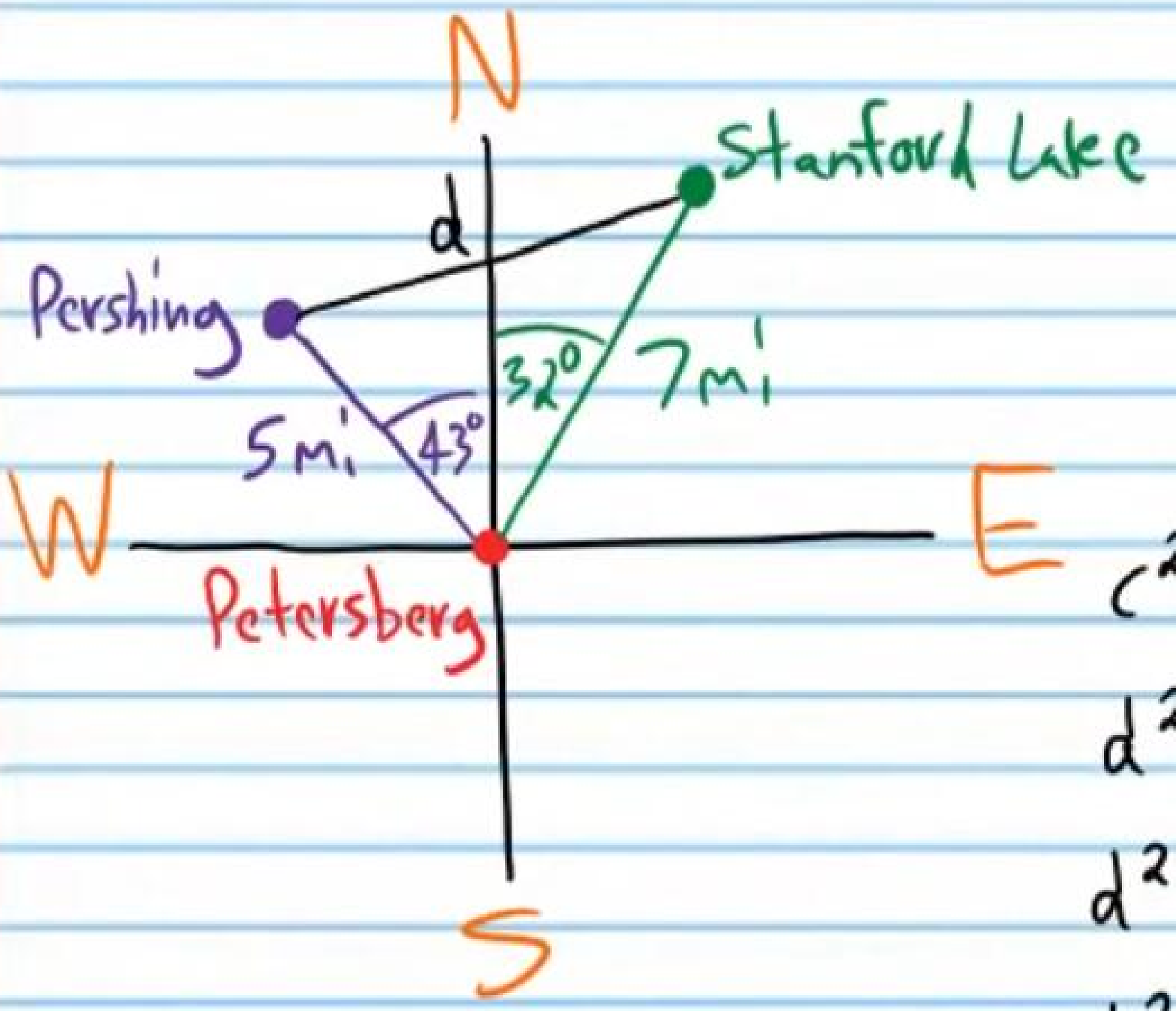
$$d^2 \approx 424 - 254.558\ldots$$

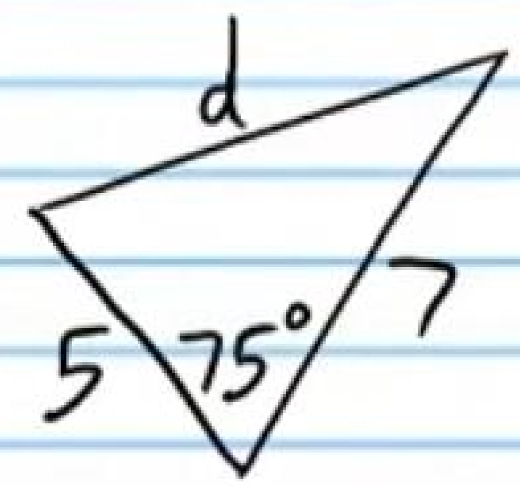
$$d^2 \approx 169.441\ldots$$

$$d \approx \sqrt{169.441\ldots}$$

$$\boxed{d \approx 13.02 \text{ km}}$$

(19) A map shows that a person in the city of Petersberg must hike 7 miles on a heading 32 degrees east of north (N32°E) to arrive at Stanford Lake. If a person wants to hike from Petersberg to the town of Pershing, he must walk 5 miles on a heading 43 degrees west of north (N43°W). How far apart are Stanford Lake and the town of Pershing?





$$c^2=a^2+b^2-2ab\cos C$$

$$d^2=5^2+7^2-2(5)(7)\cos 75^\circ$$

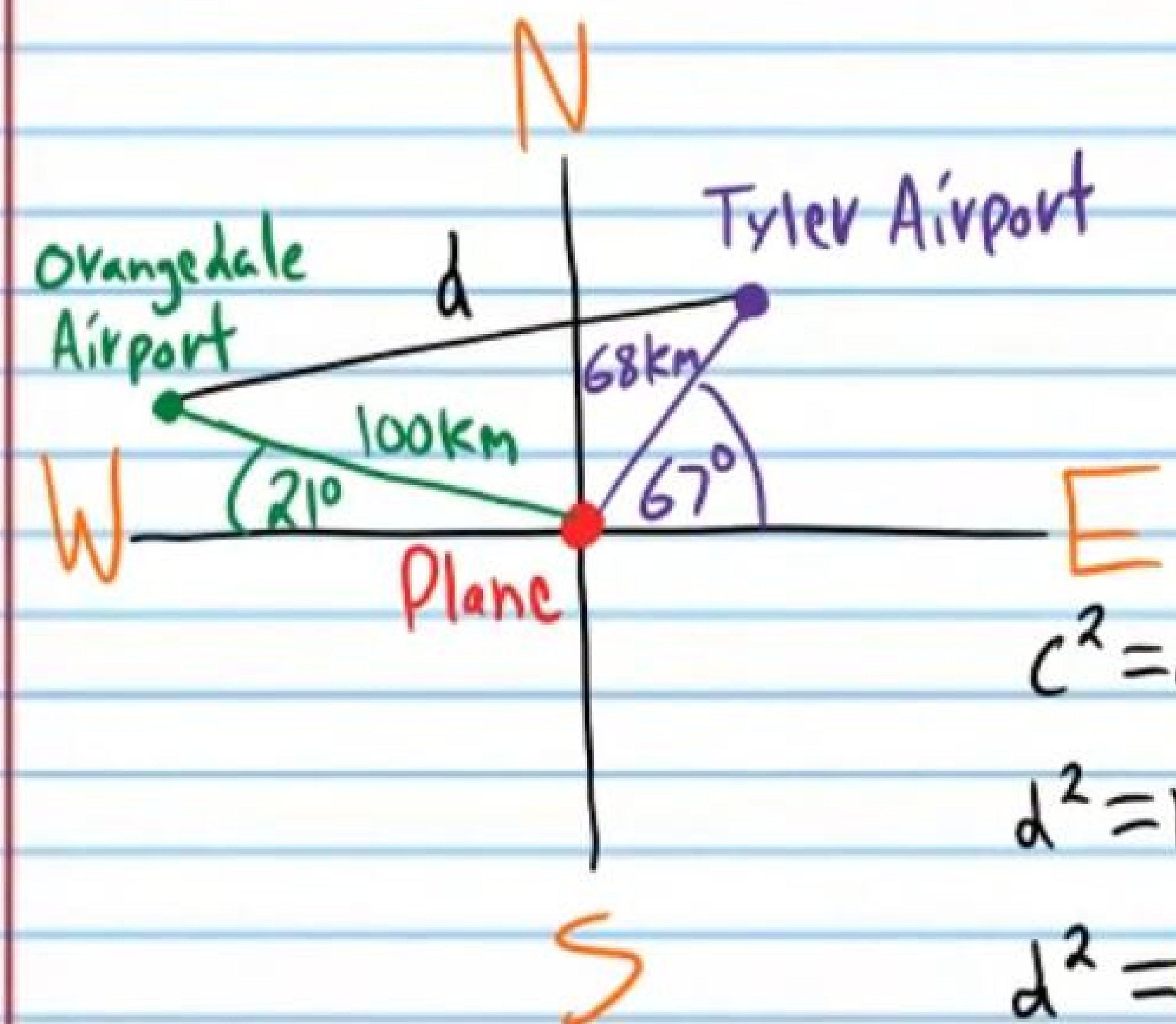
$$d^2=25+49-70\cos 75^\circ$$

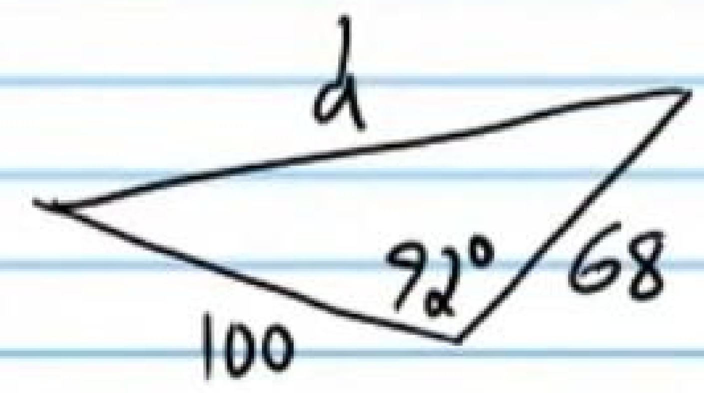
$$d^2\approx 74-18.117\ldots$$

$$d^2\approx 55.882\ldots$$

$$\boxed{d\approx 7.48\text{ mi}}$$

(20) A gps device (global positioning system) shows that a plane must fly 100 kilometers on a bearing 21 degrees north of west (W21°N) to get to Orangedale Airport. The device also shows that the same plane must fly 68 kilometers on a bearing 67 degrees north of east (E67°N) to get to Tyler Airport. How far apart are the two airports?





$$c^2=a^2+b^2-2ab\cos C$$

$$d^2=100^2+68^2-2(100)(68)\cos 92^\circ$$

$$d^2=10{,}000+4{,}624-13{,}600\cos 92^\circ$$

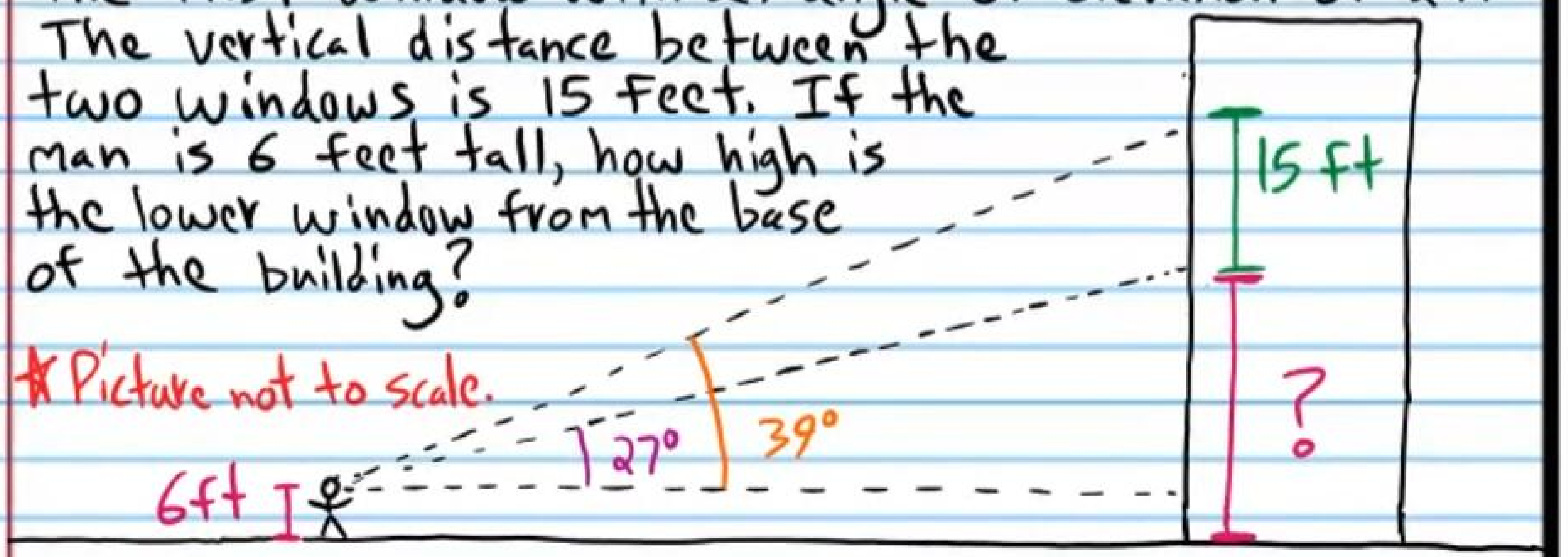
$$d^2\approx 14{,}624-(-474.633\ldots)$$

$$d^2\approx 15{,}098.633\ldots$$

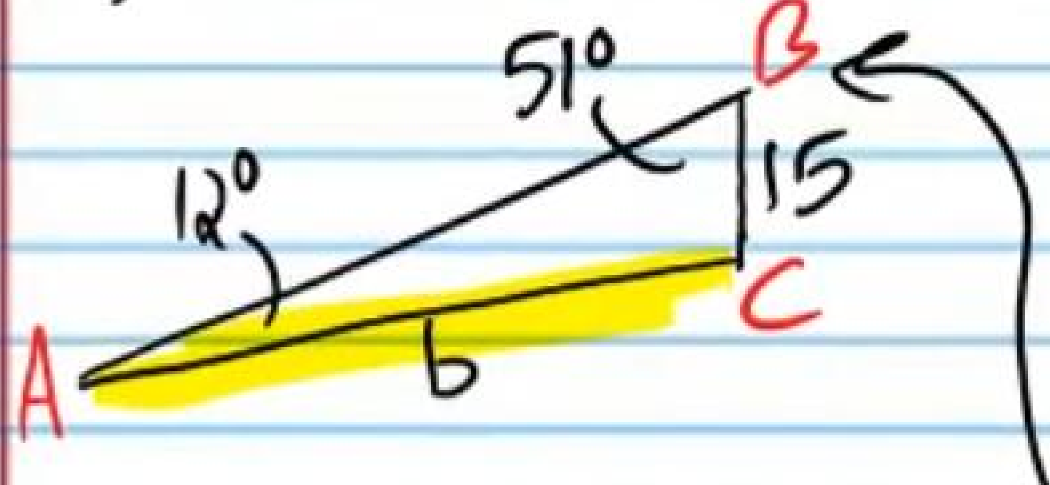
$$\boxed{d\approx 122.88\text{ km}}$$

Challenge Problems

(21) A man looks up at a window with an angle of elevation of 39°. He then looks at a window below the first window with an angle of elevation of 27°. The vertical distance between the two windows is 15 feet. If the man is 6 feet tall, how high is the lower window from the base of the building?



I) $m\angle A = 39° - 27° = 12°$



AAS

$$\frac{\sin A}{a} = \frac{\sin B}{b}$$

$$\frac{\sin 12°}{15} = \frac{\sin 51°}{b}$$

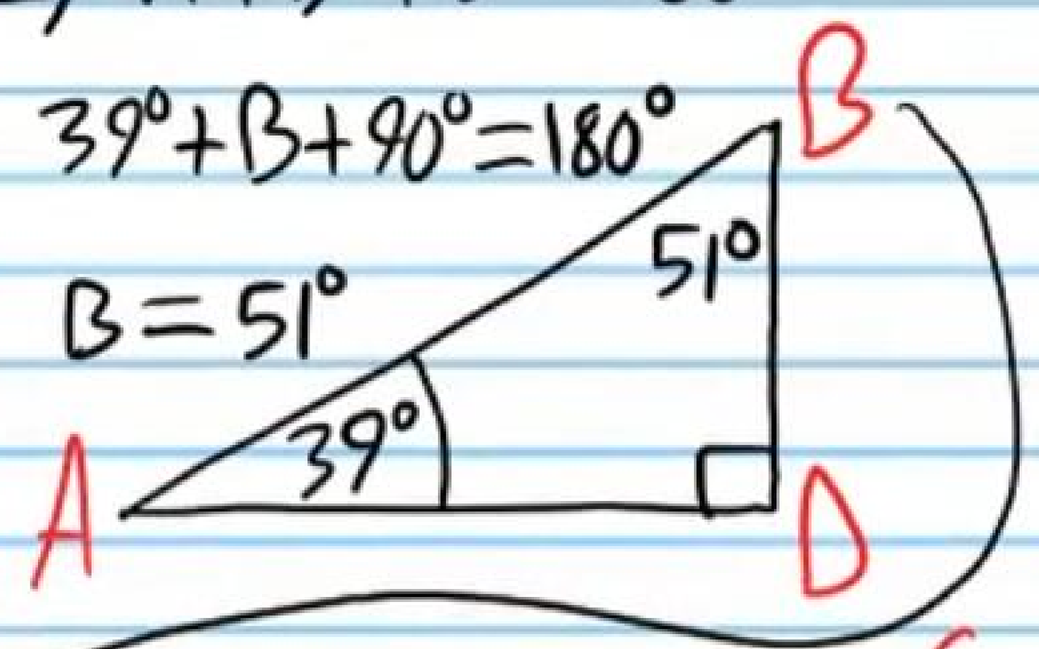
$$b = \frac{15\sin 51°}{\sin 12°}$$

$$b \approx 56.067...$$

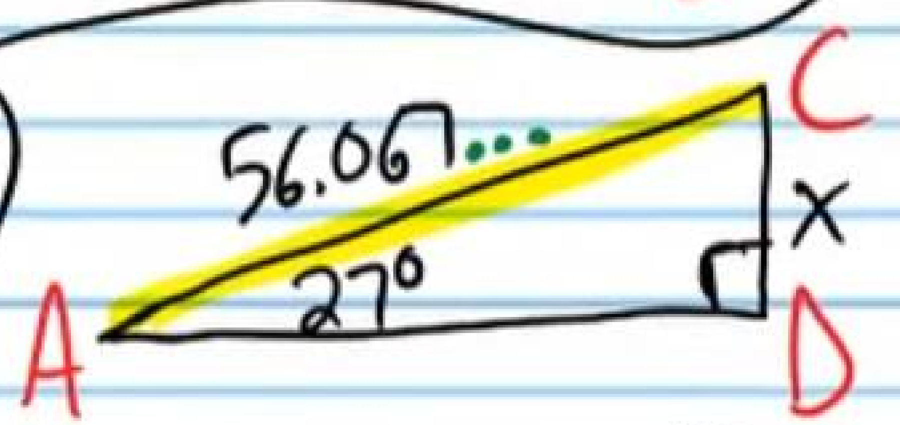
II) $A + B + D = 180°$

$39° + B + 90° = 180°$

$B = 51°$



III)

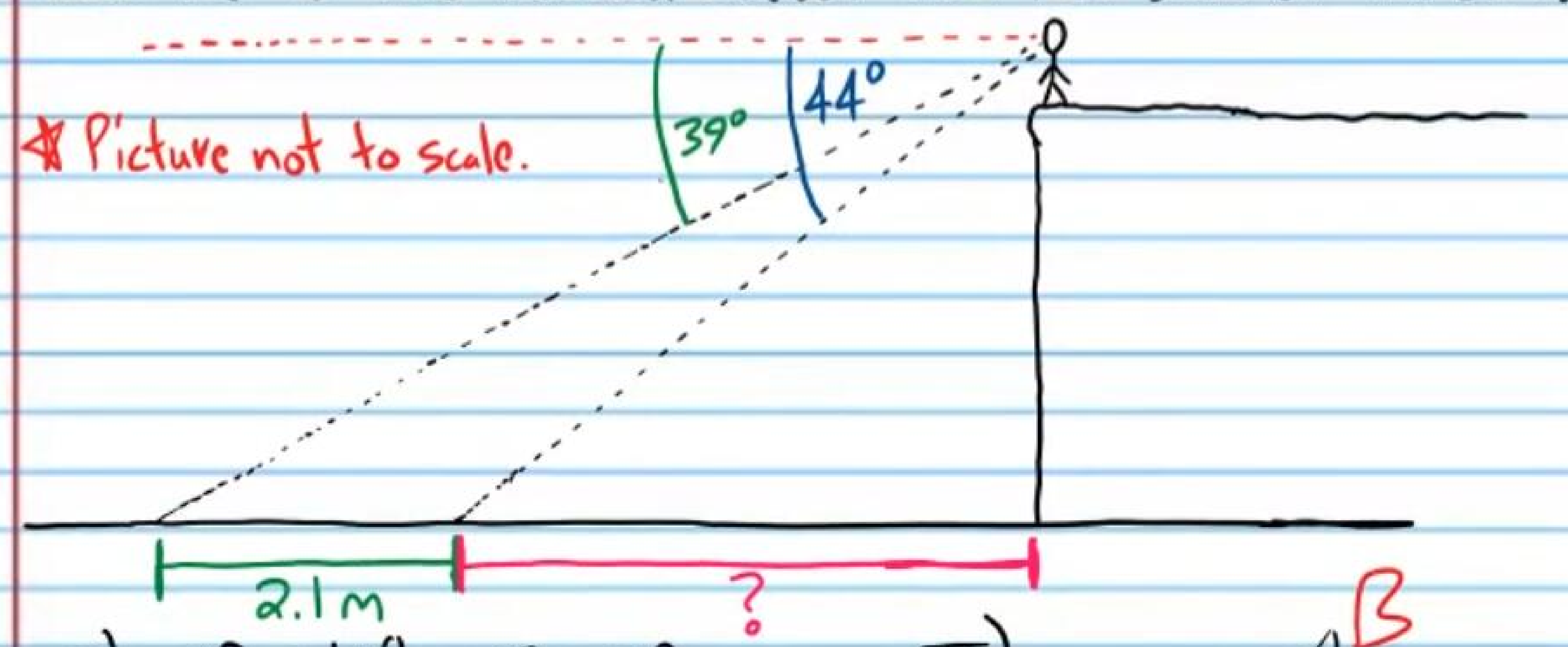


$$\sin 27° = \frac{x}{56.067...}$$

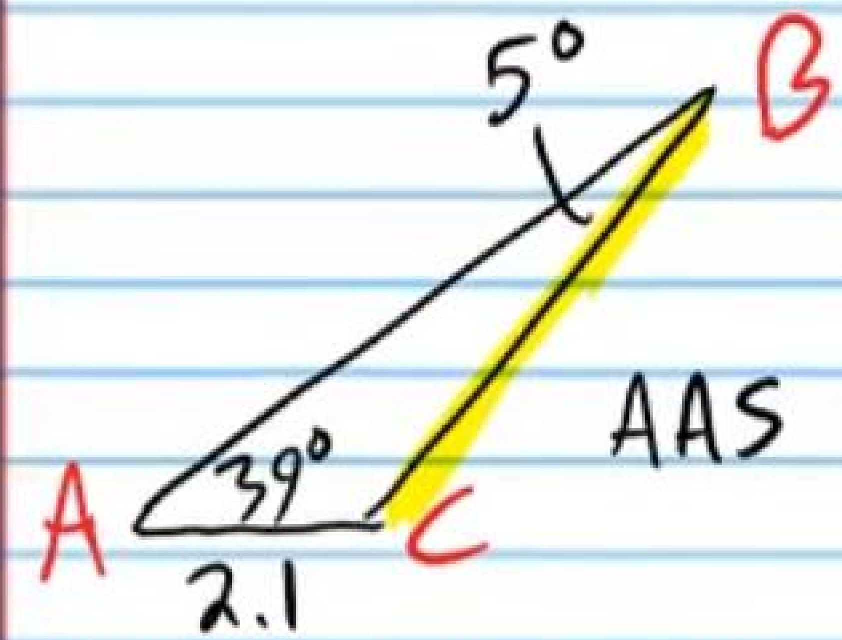
$$x \approx 25.454...$$

$$h \approx 25.454... + 6 \approx \boxed{31.45 \text{ ft}}$$

(22) A person stands on a cliff. He looks down with an angle of depression of 39° at an object. Then he looks at a different object with a 44° angle of depression. If the objects are 2.1 meters apart, how far is the second object from the base of the cliff?



I) $m\angle B = 44° - 39° = 5°$

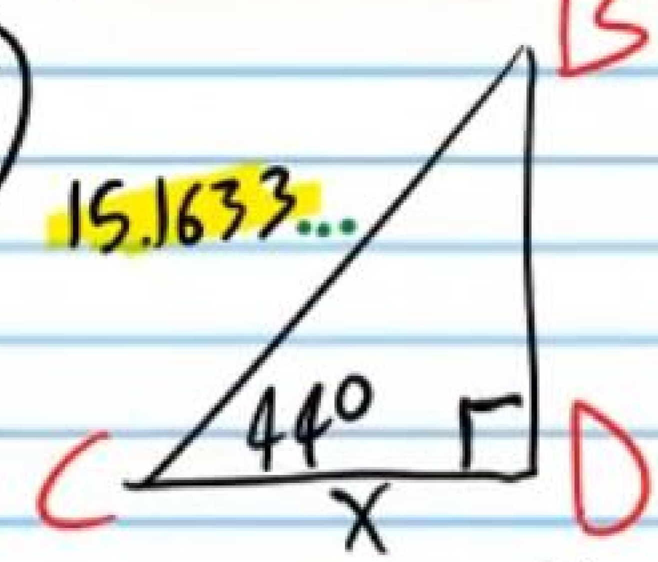


$$\frac{\sin A}{a} = \frac{\sin B}{b}$$

$$\frac{\sin 39°}{a} = \frac{\sin 5°}{2.1}$$

$$a = \frac{2.1\sin 39°}{\sin 5°} \approx 15.1633...$$

II)

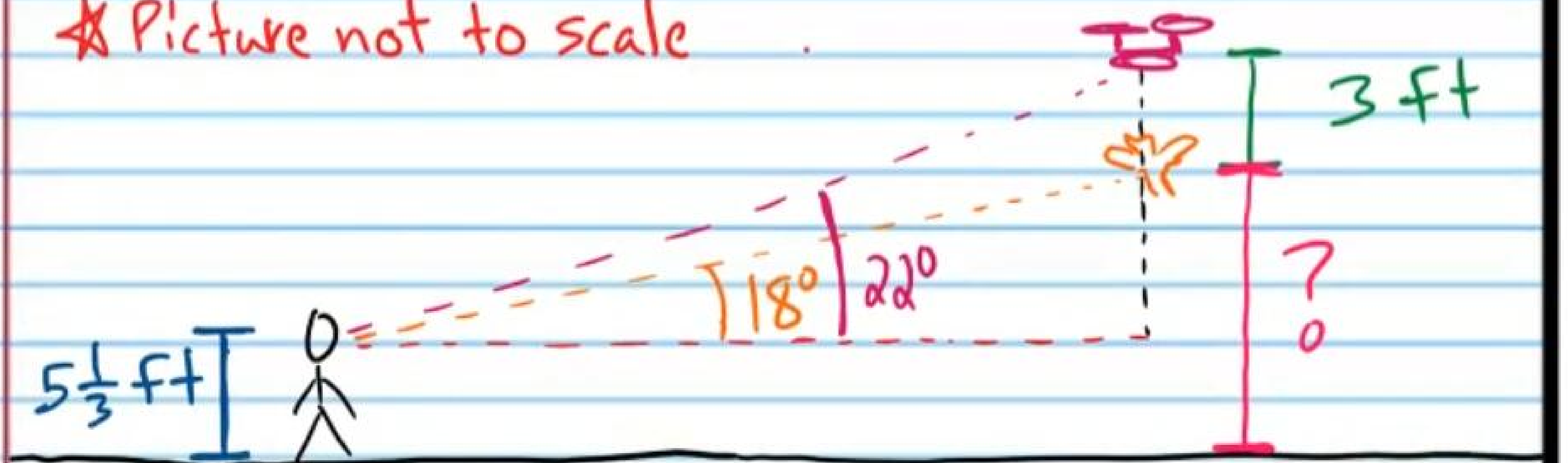


$$\cos 44° \approx \frac{x}{15.1633...}$$

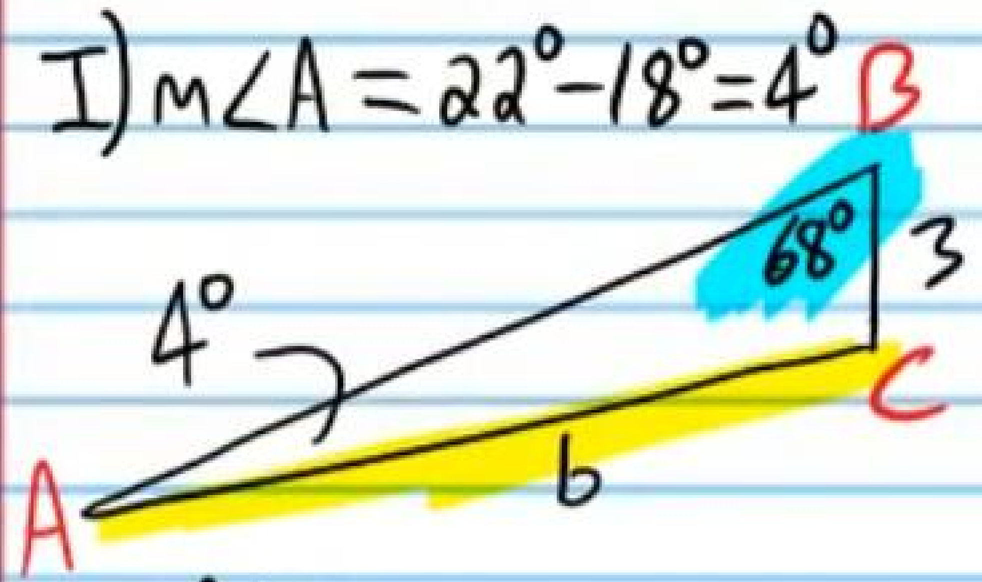
$$\boxed{x \approx 10.9 \text{ m}}$$

23. Daniel looks up at a drone hovering in the air with a 22° angle of elevation. He then looks at a bird floating in the wind directly below the drone with an 18° angle of elevation. If the bird and the drone are 3 feet apart, how high is the bird from the ground if Daniel is $5\frac{1}{3}$ feet tall?

★ Picture not to scale



I) $m\angle A = 22° - 18° = 4°$



AAS

$$\frac{\sin A}{a} = \frac{\sin B}{b}$$

$$\frac{\sin 4°}{3} = \frac{\sin 68°}{b}$$

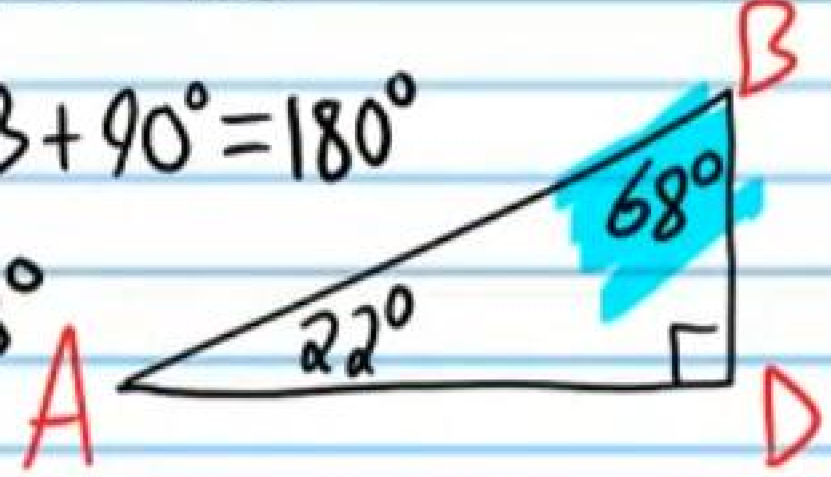
$$b = \frac{3\sin 68°}{\sin 4°}$$

$$b \approx 39.875\ldots$$

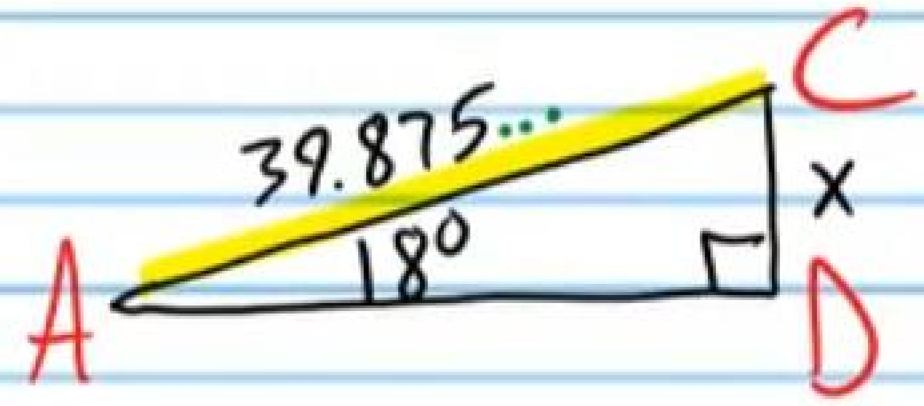
II) $A + B + D = 180°$

$$22° + B + 90° = 180°$$

$$B = 68°$$



III)



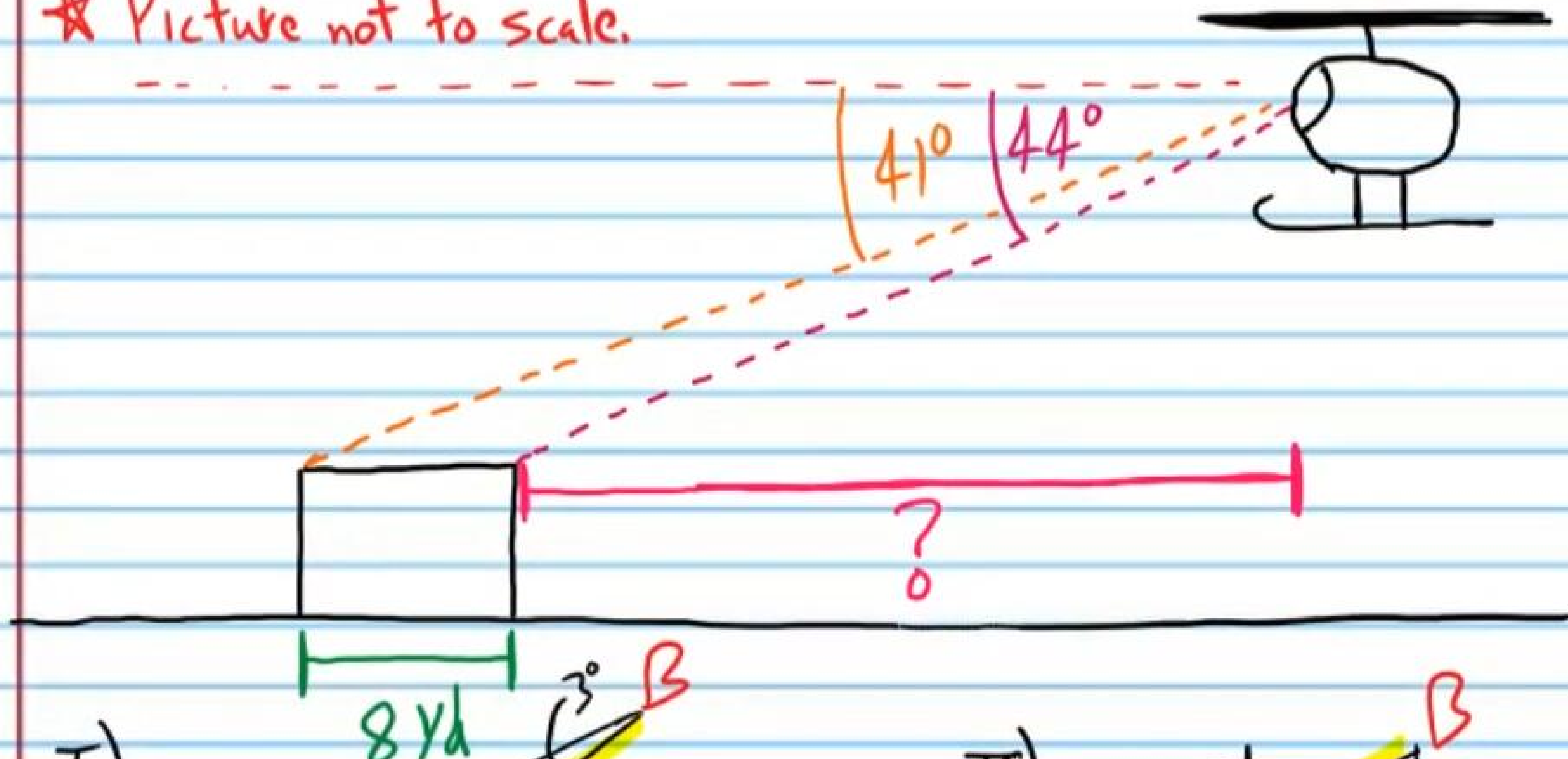
$$\sin 18° \approx \frac{x}{39.875\ldots}$$

$$x \approx 12.3221\ldots$$

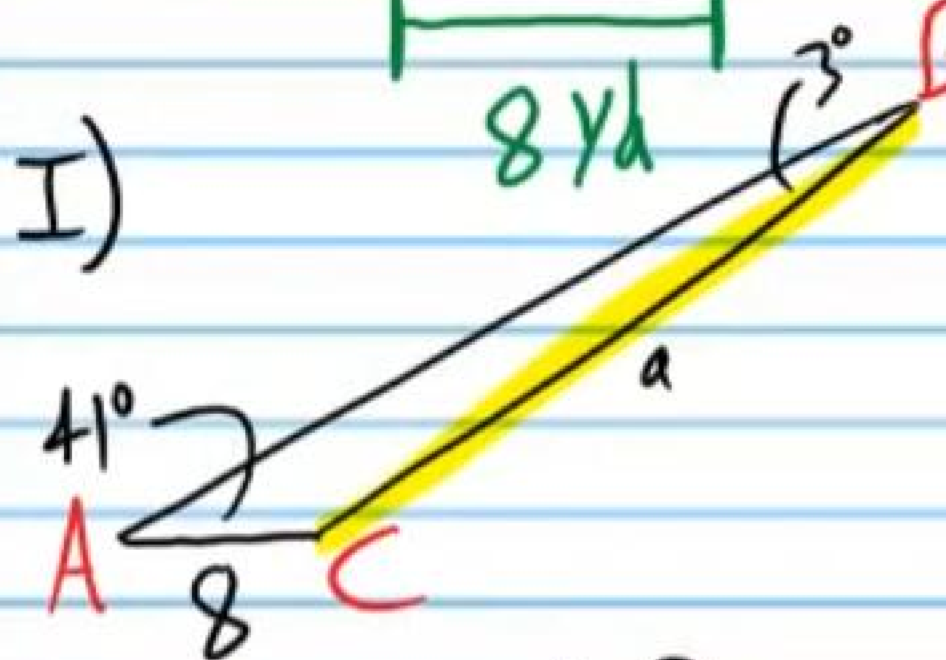
$$h \approx 12.3221\ldots + 5\tfrac{1}{3} \approx \boxed{17.66 \text{ ft}}$$

24. A helicopter will land on a platform that is 8 yards wide. The angle of depression from the cockpit to the far side of the platform is 41° and the angle of depression from the cockpit to the near side of the platform is 44°. What is the horizontal distance from the helicopter to the near side of the platform?

★ Picture not to scale.



I)



$$\frac{\sin A}{a} = \frac{\sin B}{b}$$

$$\frac{\sin 41°}{a} = \frac{\sin 3°}{8}$$

$$a = \frac{8\sin 41°}{\sin 3°} \approx 100.284\ldots$$

II)

$$\cos 44° \approx \frac{x}{100.284\ldots}$$

$$\boxed{x \approx 72.138 \text{ yd}}$$